LECTURE # 0
INTRODUCTORY LECTURE

Calculus and Analytical Geometry

- Calculus is the mathematical study of change.
- Analytical geometry, also known as Coordinate geometry or Cartesian geometry, is the study of geometry using a coordinate system.
- Geometry is the study of shape.
- Algebra is the study of operations and their application to solving equations.

➢ Study of change leads to the concept of limits, derivatives and integration.

➢ The basic concept of calculus starts from number system, which is also known as decimal system.

➢ Each number is represented by its base.

- ❖ If the base is 2, it is a binary number.
- ❖ If the base is 8, it is an octal number.
- ❖ If the base is 10, then it is called decimal number system.
- ❖ If the base is 16, it is part of the hexadecimal number system.

Relationship of calculus with computer science (in general)

- ❖ **Computer science** is the study of processes that interact with data and that can be represented as data in the form of programs.
- ❖ It is the theory, experimentation, and engineering that enables the use of algorithms to manipulate, store, and communicate digital information.
- ❖ Computers can take massive amounts of data in the form of variables and absolutes and process that data into calculations, predictions, and simulations.
- ❖ Calculus is the finding and properties of derivatives and integrals of functions.
- ❖ Mathematicians use calculus to find information about changes to a system.
- ❖ For computer science, we use calculus for general problem solving for applications and simulations. For example, it can create programs solving integrals and derivatives. In fact, calculus used on calculators were first simple computer

programs. Math is used several different ways in technology. For example, the Internet is based upon mathematical binary code.

- ❖ Computer algebra systems that compute integrals and derivatives directly, either symbolically or numerically.
- ❖ Any software that simulates a physical system that is based on continuous differential equations necessarily involves computing derivatives and integrals.

How/when is Calculus used in computer science

- Machine Learning, which makes heavy use of Statistics.
- Data Mining and related subjects, which also use lots of Statistics.
- Robotics where you will need to model physical movements of a robot, so you will need to know partial derivatives and gradients.
- Graphing and visuals.
- Applications to solve problems.
- Coding.
- Binary calculus.
- Information processing.

Relationship of calculus with computer science (in particular)

Word problems in mathematics	Computer programming
1. Comprehension, understanding the problem	
2. Devising a plan	
Determinantion of unknown variable	Declaration of variables
Mathematical model (plan)	Algorithms and their implementation (plan)
3. Carrying the plan	
Solving equations	Compiling and running
Answering to the questions of the starting problem	
4. Discussion (Looking back)	
changing initial conditions	testing
generalizing the problem	debugging
looking for other methods for solution	optimizing code

LECTURE # 1
INTERVALS

Contents:

- Number System
- Real Line
    - Line
    - Line segment
    - Ray
- Intervals
- Types of Intervals
    - Finite intervals
        - Open Intervals
        - Closed Intervals
        - Semi-open/Semi-closed Intervals
    - Infinite Intervals
- Example

INTERVALS

Number System

1. Set of natural numbers (positive integers) $= N = \{1, 2, 3, \dots\}$

- Set of prime numbers $= P = \{2, 3, 5, 7, 11, \dots\}$
- Set of composite numbers $= \{4, 6, 8, 9, 10, 12, \dots\}$

2. Set of whole numbers (non-negative integers) $= W = \{0, 1, 2, 3, \dots\}$

3. Set of integers $= Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$

- Set of even integers $= E = \{0, \pm 2, \pm 4, \pm 6, \dots\}$
- Set of odd integers $= O = \{\pm 1, \pm 3, \pm 5, \pm 7, \dots\}$

4. Set of rational numbers $= Q = \left\{x \mid x = \frac{a}{b} : a, b \in Z, b \neq 0\right\}$

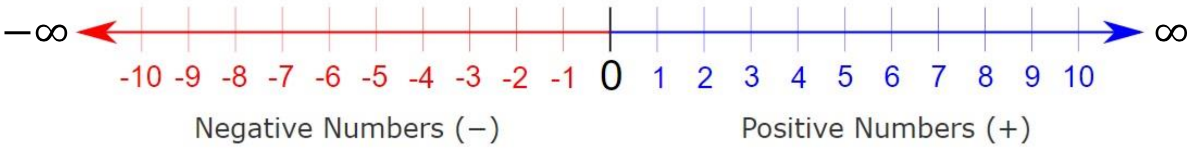
Rational numbers are formed by dividing two integers. Therefore, all integers, terminating decimals (e.g., 1.5, 3.14) and recurring decimals (e.g., $0.333\ldots = 0.\bar{3}$) are rational numbers. e.g., $\frac{3}{5}, -2, \frac{7}{2}, \sqrt{4}$ etc.

5. Set of irrational numbers $= Q' = \left\{x \mid x \neq \frac{a}{b} : a, b \in Z, b \neq 0\right\}$

Numbers that are not rational are called irrational numbers. All non-terminating, non-recurring decimals are irrational numbers. e.g., $\frac{\sqrt{3}}{5}, \pi, e, \sqrt{2}$.

6. Set of real numbers $= R = Q \cup Q' = (-\infty, \infty)$

Infinity "∞" is a symbol used to represent very large numbers and "$-\infty$" is used to represent very small numbers as shown in the number line given below:

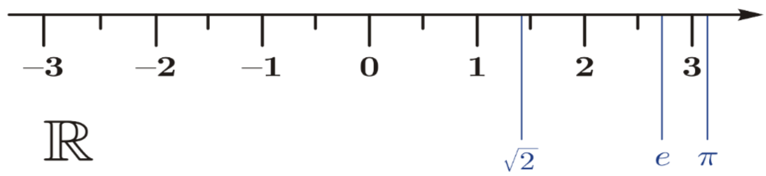


- ✓ Set of positive real numbers $= R^+ = (0, \infty)$
- ✓ If x is a positive real number, we write it as $x > 0$.
- ✓ Set of non-negative real numbers $= R^+ \cup \{0\} = [0, \infty)$
- ✓ If x is a non-negative real number, we write it as $x \geq 0$.

Calculus and Analytical Geometry
LECTURE # 1: INTERVALS

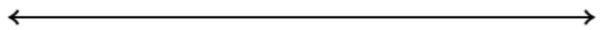
Real Line

The **real line**, or **real number line** is the line whose points are the real numbers.



Line:

A line extends forever in both directions, as follows:

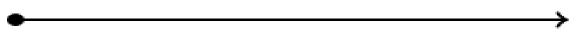


Line Segment:

A line segment is just a part (segment) of a line. It has two endpoints, as follows:

Ray:

A ray starts at one point and continues on forever in one direction, as follows:



Intervals

A subset of real line is called an interval.

OR

An interval is the set of all real numbers between two fixed numbers a and b, which are known as end points.

For example,

(i) The set of all real numbers x such that $x > 6$ is an interval, which can be written as $(6, \infty)$.

(ii) $-2 \leq x \leq 5$ is an interval which can also be written as $[-2,5]$.

Types of intervals

- **Finite intervals**
  - Closed intervals
  - Open intervals
  - Semi-open / Semi-closed Intervals
- **Infinite intervals**

Finite intervals

Interval of numbers corresponding to line segment is called a finite interval. i.e., a finite interval is an interval, whose both endpoints are numbers. There are three types of finite intervals.

(i) Closed interval

A finite interval is said to be closed interval, if it contains both of its end points.

For example, $[-5,2], [0,1], [3,7]$ *are the closed intervals.*

(ii) Open interval

A finite interval is said to be open interval, if it does not contain both end points of the interval.

For example, $(-5,2), (0,1), (3,7)$ *are the open intervals.*

(iii) Semi-open/Semi-closed interval

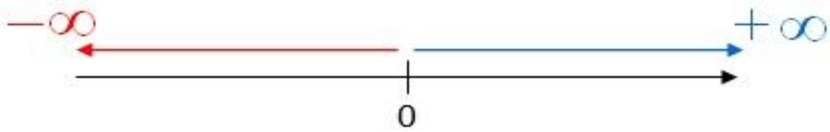
A finite interval is said to be semi-open or semi-closed interval, if it contains one end point of the interval but not the other end point.

For example, $(-5,2], [0,1), (3,7]$ *are the semi-open or semi-closed intervals.*

- ***Infinite interval***

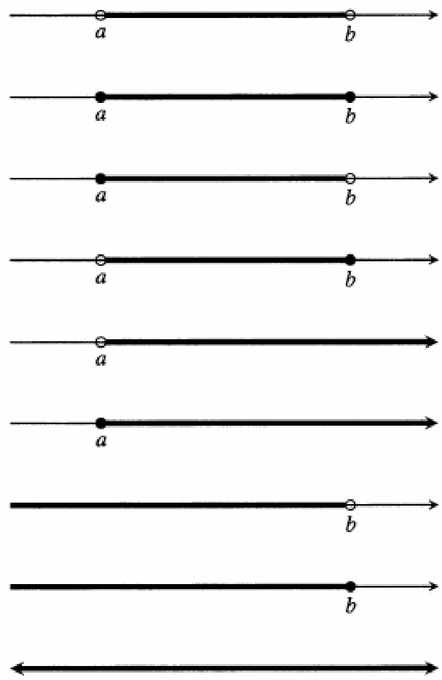
Intervals corresponding to ray and the real line are called infinite intervals.

For example, $(-\infty, 0)$ *and* $(0, \infty)$ *are infinite intervals given below:*



Let a *be the starting end point and* b *be the ending end point, then following are some examples of intervals.*

	Notation	Set	Graph
Finite:	(a, b)	$\{x \mid a < x < b\}$	
	$[a, b]$	$\{x \mid a \le x \le b\}$	
	$[a, b)$	$\{x \mid a \le x < b\}$	
	$(a, b]$	$\{x \mid a < x \le b\}$	
Infinite:	(a, ∞)	$\{x \mid x > a\}$	
	$[a, \infty)$	$\{x \mid x \ge a\}$	
	$(-\infty, b)$	$\{x \mid x < b\}$	
	$(-\infty, b]$	$\{x \mid x \le b\}$	
	$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	



Example: **Determine which of the following intervals are open interval, closed interval, semi-open/semi-closed interval?**

i. $[-2, 7]$
ii. $(-2, 0)$
iii. $(0, 2)$
iv. $(1, 4]$
v. $[0, 3)$

Solution:

i. $[-2, 7]$ is a closed interval.
ii. $(-2, 0)$ is an open interval.
iii. $(0, 2)$ is an open interval.
iv. $(1, 4]$ is a semi-open/semi-closed interval.
v. $[0, 3)$ is a semi-open/semi-closed interval.

LECTURE # 2
INEQUALITIES

Contents:

➢ Inequality:

An **inequality** is a relation between two or more algebraic expressions combined by the symbols $\neq, <, >, \leq, \geq$. For example, $4 < 5,\ x \neq 7,\ x \geq 1$ are inequalities.

- The notation $a < b$ means that a is less than b or a is strictly less than b.
- The notation $a > b$ means that a is greater than b, or a is strictly greater than b.

In either case, a is not equal to b.

- The notation $a \leq b$ means that a is less than or equal to b.
- The notation $a \geq b$ means that a is greater than or equal to b.

➢ Rules of Inequalities:

If a, b and c real numbers, then the following properties hold:

- If $a < b$, then this implies tha $a + c < b + c$.
- If $a < b$, then this implies tha $a - c < b - c$.
- If $a < b$ and $c > 0$, then this implies tha $ac < bc$.
- If $a < b$ and $c < 0$, then this implies tha $ac > bc$ or $bc < ac$.
  - If $a < b$, then this implies that $-a > -b$.
- If $a > 0$, then $\frac{1}{a} > 0$.
- If a and b are both positive or both negative such that $a < b$, then this implies that $\frac{1}{a} > \frac{1}{b}$.

➢ Solving Inequalities:

The process of finding the interval or intervals of numbers that satisfy an inequality in the given variable, is called solving the inequality.

Types of inequalities:

- Linear inequalities
- Rational inequalities
- Quadratic inequalities

Linear inequality:

The expression in which the highest power of the variable is 1, combined with the symbols $<, >, \leq, \geq$ is known as linear inequality.
For example, $-7 - x > 5$, $5x - 2 \leq 6$, $3x + 1 < 0$ are all linear inequalities.

The general form of linear inequality containing any symbol from $<$, $>$, $\leq$, $\geq$ in one variable is:

$$ax + b < 0, \quad a \neq 0$$

$$ax + b \leq 0, \quad a \neq 0$$

$$ax + b > 0, \quad a \neq 0$$

$$ax + b \geq 0, \quad a \neq 0$$

Example 1:

Solve $3x - 5 \leq 3 - x$

Solution:

Add 5 on both sides of the inequality

$$3x - 5 + 5 \leq 3 + 5 - x$$

$$3x \leq 8 - x$$

Add x on both sides.

$$3x + x \leq 8 - x + x$$

$$4x \leq 8$$

Finally, divide both sides of the inequality by 4 to get;

$$x \leq 2$$

The solution of the given inequality is the interval $(-\infty, 2]$.

Example 2:

Solve the inequality $2x - 1 < x + 3$ and graph their solution set on the real line.

Solution:

$$2x - 1 < x + 3$$

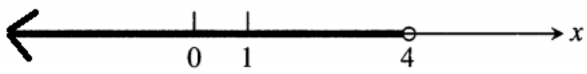
Adding 1 on both sides, we have

$$2x - 1 + 1 < x + 3 + 1$$
$$2x < x + 4$$

Subtract x from both sides, we have

$$2x - x < x + 4 - x$$
$$x < 4$$

The solution of the given inequality is $(-\infty, 4)$. Graphically, it can be expressed as



➢ Rational inequality:

An inequality, which contain a rational expression, is called rational inequality.

For example, $\frac{3}{2x} > 1$, $\frac{2x}{x-3} < 4$, $\frac{2x-3}{x-6} > x$ are all rational inequalities.

Example 3:

Solve the inequality $-\frac{x}{3} < 2x + 1$ and graph their solution set on the real line.

Solution:

The given inequality is

$$-\frac{x}{3} < 2x + 1$$

$$-\frac{x}{3} - 2x < 1$$

By taking L.C.M

$$\frac{-x - 6x}{3} < 1$$

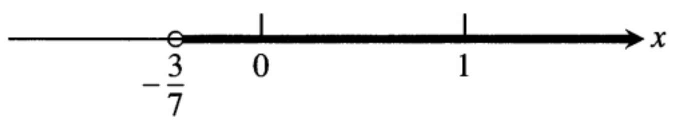
$$\frac{-7x}{3} < 1$$

Multiply by $-\frac{3}{7}$ both sides of inequality, we get

$$\left(-\frac{3}{7}\right)\left(\frac{-7x}{3}\right) > 1\left(-\frac{3}{7}\right)$$

$$x > -\frac{3}{7}$$

The solution of the given inequality is $\left(-\frac{3}{7}, +\infty\right)$. Graphically, it can be expressed as



Example 4:

Solve the inequality $\frac{6}{x-1} \geq 5$ and graph their solution set on the real line.

Solution:

The given inequality is

$$\frac{6}{x-1} \geq 5$$

Subtract 5 from both sides, we get

$$\frac{6}{x-1} - 5 \geq 5 - 5$$
$$\frac{6}{x-1} - 5 \geq 0$$

Taking L.C.M, we get

$$\frac{6 - 5(x-1)}{x-1} \geq 0$$
$$\frac{6 - 5x + 5}{x-1} \geq 0$$
$$\frac{11 - 5x}{x-1} \geq 0$$

From the above inequality, we get two equations as follows:

$$11 - 5x = 0$$

and

$$x - 1 = 0$$

By solving these two equation, we get

$$x = \frac{11}{5}, \quad x = 1$$

These two points divide the real line into three intervals:

$$(-\infty, 1), \qquad \left(1, \frac{11}{5}\right), \qquad \left(\frac{11}{5}, +\infty\right)$$

Here, we will find the solution set interval using test points.

Consider the following test point:

Test Points	Interval	Corresponding inequality	Substitution of test point in inequality	Test point (Satisfies/Not satisfies)
$x = 0$	$(-\infty, 1)$	$\frac{11-5x}{x-1} \geq 0$	$-11 \geq 0$ (which is not possible)	Not satisfies the inequality
$x = 2$	$\left(1, \frac{11}{5}\right)$	$\frac{11-5x}{x-1} \geq 0$	$1 \geq 0$ (which is possible)	Satisfies the inequality
$x = 3$	$\left(\frac{11}{5}, +\infty\right)$	$\frac{11-5x}{x-1} \geq 0$	$-2 \geq 0$ (which is not possible)	Not satisfies the inequality

From these three intervals $(-\infty, 1)$, $\left(1, \frac{11}{5}\right)$, $\left(\frac{11}{5}, +\infty\right)$, one interval $\left(1, \frac{11}{5}\right)$ is satisfied the condition. Note that, the boundary point $x = \frac{11}{5}$, satisfied the inequality. Hence, the solution set of the given inequality is $(1, \frac{11}{5}]$. The graph of the solution set of the inequality is given as follows:

Quadratic inequality:

The expression in which the highest power of the variable is 2, combined with the symbols $<$, $>$, $\leq$, $\geq$ is known as quadratic inequality. For example,

$x^2 - x > 5, x^2 + 5x < 6, 2x^2 - 3x + 1 < 0$ are all quadratic inequalities.

The general form of quadratic inequality containing any symbol from $<$, $>$, $\leq$, $\geq$ in one variable is:

$$ax^2 + bx + c < 0, \quad a \neq 0$$
$$ax^2 + bx + c \leq 0, \quad a \neq 0$$
$$ax^2 + bx + c > 0, \quad a \neq 0$$
$$ax^2 + bx + c \geq 0, \quad a \neq 0$$

Example 5:

Find all those numbers whose squares are greater or equal to two less than three times the number.

Solution:

Let x be the number that satisfied the condition. Then the inequality is:

$$x^2 \geq 3x - 2$$

We can write is as

$$x^2 - 3x + 2 \geq 0$$
$$x^2 - 2x - x + 2 \geq 0$$

$$x(x-2)-1(x-2)\geq 0$$
$$(x-2)(x-1)\geq 0$$

From the above inequality, we get two equations as follows:

$$x-2=0,\quad x-1=0$$

This implies that

$$x=2,\qquad x=1$$

We get boundary points $x=1$ and $x=2$, which divides the real line into three intervals as follows:

$$(-\infty,1),\qquad (1,2),\qquad (2,+\infty)$$

Here, we will find the solution set interval using test points.

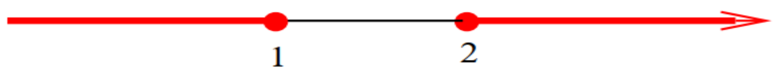
Consider the following test point:

Test Points	Interval	Corresponding inequality	Substitution of test point in inequality	Test point (Satisfies/Not satisfies)
$x=0$	$(-\infty,1)$	$(x-2)(x-1)\geq 0$	$2\geq 0$ (which is possible)	Satisfies the inequality
$x=1.5$	$(1,2)$	$(x-2)(x-1)\geq 0$	$-2.5\geq 0$ (which is not possible)	Not satisfies the inequality
$x=3$	$(2,+\infty)$	$(x-2)(x-1)\geq 0$	$2\geq 0$ (which is possible)	Satisfies the inequality

From these three intervals $(-\infty,1)$, $(1,2)$, $(2,+\infty)$, one interval $(1,2)$ is not satisfied the inequality while two intervals $(-\infty,1)$ and $(2,+\infty)$ are satisfying the inequality.

Note that boundary point $x=1$ and $x=2$ are satisfying the inequality, hence are the part of solution set.

Hence, the solution set of the given inequality is $(-\infty,1]\cup[2,+\infty)$. The graph of the solution set of the inequality is given as follows:



Practice questions:

Find the solution set of the following inequalities and graph their solution set.

- $5x - 3 \leq 7 - 3x$
- $\frac{4}{5}(x-2) < \frac{1}{3}(x-6)$
- $\frac{-2}{x-1} \leq -x$
- $\frac{1}{x+2} < \frac{2}{x-3}$
- $x^2 > x + 2$
- Find all numbers whose squares are less or equal to six more than the numbers.

Graph of an Equation

Cartesian Coordinate System

A Cartesian coordinate system consists of two perpendicular number lines that intersect at their origins (on a plane).

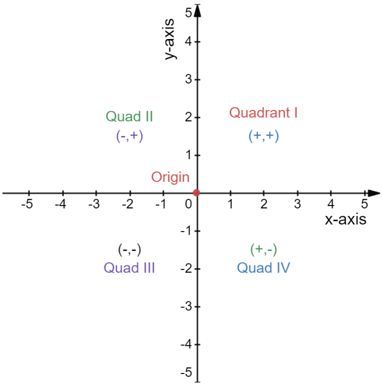
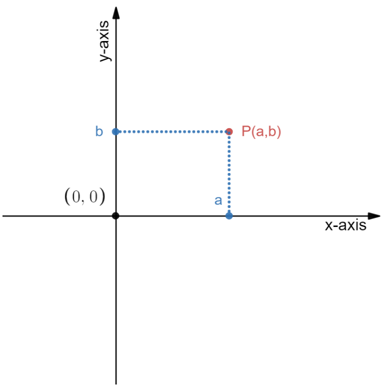


FIGURE 1. xy-coordinate system

- The lines are called coordinate axes.
- Generally, one line is horizontal, called the x-axis, and the other is vertical, called the y-axis.
- The horizontal line towards the right of the origin is called the positive x-axis.
- The horizontal line towards the left of the origin is called the negative x-axis.
- The vertical line above the origin is called the positive y-axis.
- The vertical line below the origin is called the negative y-axis.
- The plane on which a coordinate system is drawn is called the xy-plane.
- The axes divide the coordinate plane into four parts, called quadrants, and are numbered I, II, III, and IV.

Point verses Ordered Pair

- To each point P of the xy-plane there corresponds an ordered pair (a, b), see the attached figure.



- a is called the x-coordinate of P; b is called the y-coordinate of P.

- a is actually the distance of P from the y-axis; b is the distance of P from the x-axis.
- To each ordered pair (a, b) there corresponds a point P on the xy-plane, called the graph of the ordered pair.

Graph of an Equation

The graph of an equation is the graph of all the ordered pairs (a, b) that satisfy the equation.

- Since there are usually an infinite number of ordered pairs that satisfy an equation, a sketch of the graph is generally sufficient.
- A simple approach to sketch a graph is to plot some reasonable number of ordered pairs, then connect them with a smooth curve or line.

Sketching the Graph

Sketching the graph of an equation is an 8-step process. Let's to understand this process in the following example.

Example 1. Sketch the graph of the equation $y = \frac{-x^3}{6} + 2x + 5$.
Solution.
Step 1. (Find an interval for sketching) Find an interval on which the graph will be sketched. If it is given, it is fine; otherwise, determine it yourself. If you face difficulty, take the interval $[-5, 5]$; it works most of the times. Since, in the present example, the interval is not given, we take the interval $[-5, 5]$.

Step 2. (Find x-values) In order to sketch a graph, we need both the x-values and the y-values. The x-values come from the interval, and the y-values come from the given equation. Take integer values of x from the interval. For instance, for the present example we take $x = -5, -4, -3, -1, 0, 1, 2, 3, 4, 5$.

Step 3. (Find y-values) Find y-values. The y-values will be obtained by putting the x-values in the given equation. For instance, if we take $x = -5$, then

$$\begin{aligned} y &= \frac{-(-5)^3}{6} + 2(-5) + 5 \\ &= \frac{-(-125)}{6} - 10 + 5 \\ &= \frac{125}{6} - 5 \approx 15.8 \end{aligned}$$

Similarly, find all other y-values, and write these x- and y-values in a table:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y	15.8	7.6	3.5	2.3	3.1	5	6.8	7.6	6.5	2.3	-5.8

These x- and y-values in the form of ordered pairs are:

$(-5, 15.8), (-4, 7.6), (-3, 3.5), (-2, 2.3), (-1, 3.1)$,$(0, 5), (1, 6.8), (2, 7.6)$, $(3, 6.5)$, $(4, 2.3), (5, 5.8)$

Step 4. (Draw the xy-plane) Draw the xy-plane at the center of the page. Both the x- and y-axes should be 12 cm long.

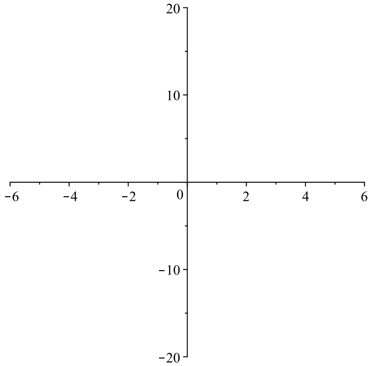


FIGURE 2. xy-plane

Step 5. (Choose a horizontal scale) Choose the horizontal scale. For this, find the span (the difference of the largest and smallest values) of the x-values. In our case, it is 5-(-5)=10. Since we (always) want to fit our graph within 10 centimeters, we divide the span by 10. This gives 1. It follows that the horizontal scale is: 1 cm = 1 unit. That is, we will mark point 1 at 1 cm on the right of the the origin, point 2 at 2 cm on the right of the the origin, and so on. Similarly, we will mark -1 at 1 cm on the left of the origin, -2 at 2 cm on the left of the origin, and so on. Thus, all the x-values will appear at 1 cm from each other. These points are marked in Figure 2. (I must mention here that this graph is produced with a graphing tool, MAPLE. So, the points may not look exactly at 1 cm from each other.)

Step 6. (Choose a vertical scale) Choose the vertical scale. Find the span of the y-values, which is 15.8-(-5.8)=21.6. Here as well, we will fit y-values within 10 centimeters. So,we divide the span by 10. This gives 2.16. It guides us to choose the horizontal scale as: 1 cm = 2.5 units. (Of course, we can take 1 cm = 2 units.) That is, we will mark point 2.5 at 1 cm vertically up from the origin, the point 5 at 2 cm vertically up from the origin, and so on. Similarly, we go for negative points. Thus, the y-points will appear at a difference of 2.5 from each other. You can see these points in Figure 2, though all y-values are not visible here; this is again due to default setting of MAPLE.

Step 7. (Mark the ordered pairs) Mark the ordered pairs (obtained in Step 3) in the xy-plane, taking care of x- and y-values.

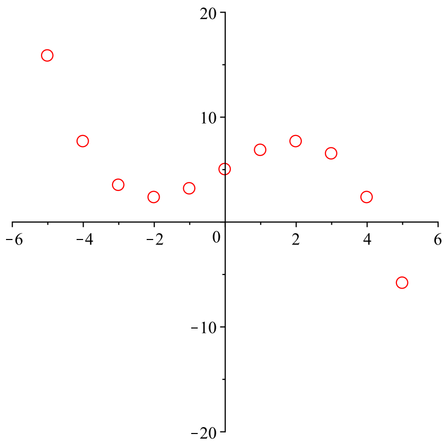


FIGURE 3. Ordered points are marked.

Step 8. (Sketch the graph) Join the marked points with a smooth curve. You must join the points taking care of the sequence of x-values; this will help you avoiding abrupt changes in the graph.

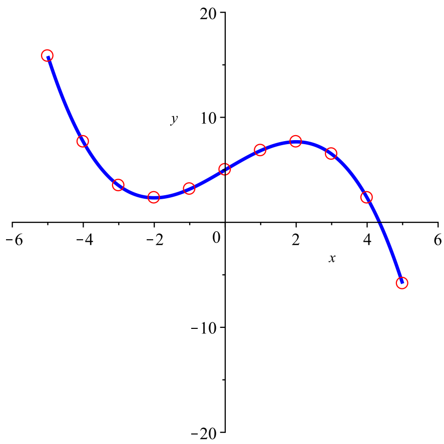


FIGURE 4. Graph of $y = \frac{-x^3}{6} + 2x + 5$

$\square$

Example 2. Sketch the graph of the equation $y = x^2$ over the interval $[-3, 3]$.
Solution.
Step 1. (Table of values)

x		-3	-2	-1	0	1	2	3
y		9	4	1	0	1	4	9

Step 2. (Scale)

- Horizontal scale: Span $= 3-(-3) = 6$, Scale $= \dfrac{\text{span}}{\text{no. of x-values}} = \frac{6}{6} = 1$
- Vertical scale: Span $= 9-(0) = 9$, Scale $= \dfrac{\text{span}}{\text{no. of y-values}} = \frac{9}{5} = 1.8 \approx 2$

Step 3. (Sketch)

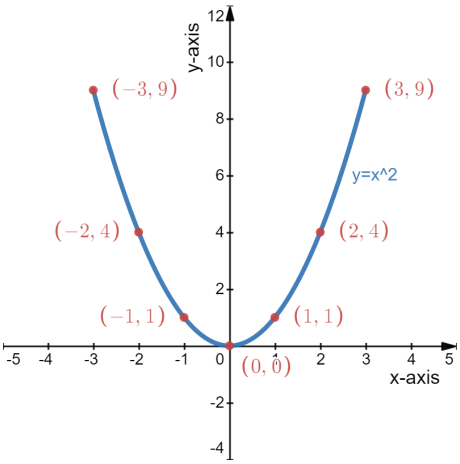


FIGURE 5. Graph of $y = x^2$

$\square$

Practice Problems

1. $y = -3$
2. $y = 2x + 3$
3. $y = -x^2 + 2$
4. $y = -x^3, [-3, 3]$
5. $y = \frac{1}{x}, [-5, 5]$
6. $y = \frac{1}{x^2}, [-5, 5]$